

第20讲. 曲线积分与路径无关条件. 全微分方程.

作业: P166. 1. 2. 3. 5. 7. 8. 13. 14 ①. ②. 15.

$$\oint_{\partial D^+} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = c.$$

$$Q = x, \quad P = -y.$$

$$\oint_{\partial D^+} x dy - y dx = 2 \iint_D dx dy = 2 S_D.$$

$$\therefore S_D = \frac{1}{2} \oint_{\partial D^+} x dy - y dx.$$

例1. 求 $D = \left\{ (x, y) \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}$ 面积.

$$\begin{aligned} \text{解: } S_D &= \frac{1}{2} \oint_{\partial D^+} x dy - y dx \\ &= \frac{1}{2} \int_0^{2\pi} (a b \cos^2 \theta + a b \sin^2 \theta) d\theta \\ &= \pi ab. \end{aligned}$$

$$\partial D^+: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \text{ 逆时针}$$

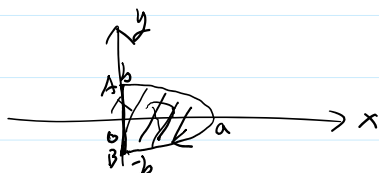
$$\begin{cases} x = a \cos \theta \\ y = b \sin \theta \end{cases}$$

$$\theta: 0 \rightarrow 2\pi.$$

例2. 计算 $I = \int_{\partial D^+} (y^2 - y \sin 2x) dx + \cos^2 x dy$.

$$\partial D^+: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ 右半部分. } (x \geq 0). \text{ 顺时针.}$$

解:



$$P(x, y) = y^2 - y \sin 2x, \quad Q(x, y) = \cos^2 x.$$

$$\frac{\partial Q}{\partial x} = -\sin 2x, \quad \frac{\partial P}{\partial y} = 2y - \sin 2x.$$

$$\therefore - \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \iint_D 2y dx dy = 0$$

$$\oint_{\overline{BA} \cup \partial D^+} P dx + Q dy =$$

$$\gamma_{\overline{BA} \cup \ell^+}^{Pdx + Qdy} =$$

$$\begin{aligned} \therefore I &= \int_{\ell^+} Pdx + Qdy = \int_{\overline{AB}} Pdx + Qdy \\ &= \int_b^{-b} Q(0, y) dy = \int_b^{-b} dy = -2b. \end{aligned}$$

$$\overline{AB}: x=0, y: b \rightarrow -b$$

$$\oint_{\ell^+} Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

$$\begin{aligned} &\parallel \\ &\oint_{\ell^+} (P, Q) \cdot \vec{c} ds \\ &\quad \downarrow \text{环量} \end{aligned}$$

$$(\cos \alpha + i \cos \beta) \cdot e^{-\frac{\pi}{2}i}$$

$$= (\cos \alpha + i \cos \beta) (-i)$$

$$= (\cos \beta - i \cos \alpha)$$

$$\vec{n} = (\cos \beta, -\cos \alpha)$$

$$\oint_{\ell^+} (P, Q) \cdot \vec{n} ds = \oint_{\ell^+} (P \cos \beta - Q \cos \alpha) ds.$$

$\downarrow$
通量

$$= \oint_{\ell^+} P dy - Q dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy.$$

$\vec{n}$ — 外法向量.



$$\vec{c} = (\cos \alpha, \cos \beta)$$

$$\vec{c} ds = (dx, dy)$$

$$\begin{cases} dx = \cos \alpha ds \\ dy = \cos \beta ds \end{cases}$$



= . 曲线积分与积分路径无关条件.

定义1. 若区域 $D \subset \mathbb{R}^2$ 中任一简单闭曲线均全在 D 中. 称 D 单连通, 否则多连通.



多连通.

定理1. 设 $D \subset \mathbb{R}^2$ 单连通区域. 且 $P(x, y), Q(x, y) \in C(D)$. 则下列等价:

①. $\int_{\ell(A, B)} Pdx + Qdy$ 与积分路径无关, 只与起点 A 和终点 B 有关.

$$(2) \exists u(x, y) \in C^1(D) \text{ s.t. } du = p dx + q dy.$$

$$(3) \forall (x, y) \in D. \quad \frac{\partial p}{\partial y}(x, y) = \frac{\partial q}{\partial x}(x, y).$$

$$(4) \text{ 对于 } D \text{ 内任意光滑或分段光滑封闭曲线 } \gamma, \quad \oint_{\gamma} p dx + q dy = 0.$$

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

$$\text{证: } (1) \Rightarrow (2).$$

$$A(x_0, y_0).$$

$$B(x, y)$$

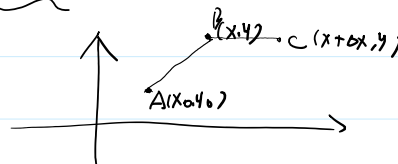
$$u(x, y) = \int_{(x_0, y_0)}^{(x, y)} p(v, w) dv + q(v, w) dw$$

$$\text{证: } \forall (x, y) \in D.$$

$$\frac{\partial u}{\partial x}(x, y) = p(x, y)$$

$$\frac{\partial u}{\partial y}(x, y) = q(x, y).$$

$$\frac{\partial u}{\partial x}(x, y) = \lim_{\Delta x \rightarrow 0} \frac{u(x+\Delta x, y) - u(x, y)}{\Delta x}$$



$$u(x+\Delta x, y) - u(x, y) = \int_{(x_0, y_0)}^{(x+\Delta x, y)} p(v, w) dv + q(v, w) dw - \int_{(x_0, y_0)}^{(x, y)} p dv + q dw$$

$$= \int_{(x, y)}^{(x+\Delta x, y)} p(v, w) dv + \underbrace{q(v, w) dw}_{=0}$$

$$= \int_x^{x+\Delta x} p(v, y) dv \stackrel{\exists \theta \in (0, 1)}{=} p(x+\theta \Delta x, y) \Delta x$$

$$\lim_{\Delta x \rightarrow 0} \frac{u(x+\Delta x, y) - u(x, y)}{\Delta x} = \lim_{\Delta x \rightarrow 0} p(x+\theta \Delta x, y) = p(x, y)$$

$$\parallel$$

$$\frac{\partial u}{\partial x}(x, y)$$

$$(2) \Rightarrow (3).$$

$$du = p dx + q dy$$

$$\frac{\partial u}{\partial x} = p.$$

$$\frac{\partial u}{\partial y} = q.$$

$$\frac{\partial p}{\partial y} = \frac{\partial^2 u}{\partial y \partial x} \in C$$

$$\frac{\frac{\partial u}{\partial y}}{\frac{\partial u}{\partial x}} = \frac{\frac{\partial^2 u}{\partial y \partial x}}{\frac{\partial^2 u}{\partial x \partial y}} \in \mathbb{C}$$

$$\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x \partial y} \in \mathbb{C}$$

(3) $\Rightarrow$ (1).

$\forall (x, y) \in D$.

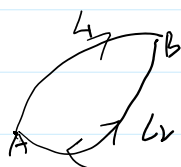
$$\frac{\partial p}{\partial y} = \frac{\partial q}{\partial x}.$$

格林公式



D 单连通

(4) $\Rightarrow$ (1).



$$\int_{L(A, B)} p dx + q dy = \int_{L(B, A)} p dx + q dy$$

$$\oint_{L(A, B) + L(B, A)} p dx + q dy = 0.$$



①. ②. ④. 对任意, $D \subset \mathbb{R}^2$ (有界闭区域) 均作.

定义2. 设 $D \subset \mathbb{R}^2$ 有界闭区域. $p(x, y), q(x, y) \in C^1(D)$. 若 $\exists u(x, y) \in C^1(D)$,

s.t. $du = p dx + q dy$. 则称 $u(x, y)$ 是 $p dx + q dy$ 的原函数.

定理2. 设 $D \subset \mathbb{R}^2$ 有界闭区域. 若 $p(x, y), q(x, y) \in C^1(D)$. 且 $\exists u(x, y) \in C^1(D)$ s.t. $u(x, y)$ 是

$p dx + q dy$ 的原函数. 则 $\forall A(x_1, y_1) \in D, B(x_2, y_2) \in D$.

$$\int_{L(A, B)} p dx + q dy = u(x_2, y_2) - u(x_1, y_1)$$

$$\int_a^b f(x) dx = F(x) \Big|_a^b$$

证: 考虑 $L(A, B)$: $\begin{cases} x = x(t) \\ y = y(t) \end{cases}$

A 对应 $t = \alpha$.

B ... $t = \beta$.

$$du = p dx + q dy \Rightarrow p(x, y) = \frac{\partial u}{\partial x}(x, y), \quad q(x, y) = \frac{\partial u}{\partial y}(x, y)$$

$$\int_{L(A, B)} p dx + q dy = \int_{\alpha}^{\beta} [p(x(t), y(t)) x'(t) + q(x(t), y(t)) y'(t)] dt$$

$$= \int_{\alpha}^{\beta} \frac{du(x(t), y(t))}{dt} dt$$

$$= u(x(t), y(t)) \Big|_{\alpha}^{\beta} = u(x_2, y_2) - u(x_1, y_1)$$

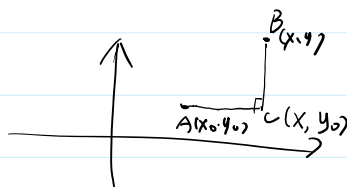
$$= u(x_2, y_2) - u(x_1, y_1)$$

三. 求 $Pdx + Qdy$ 原函数

$$D \subset \mathbb{R}^2 \text{ 单连通. } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

$$\Updownarrow \\ du = \underline{Pdx + Qdy}$$

1°. 曲线积分法.



$$A(x_0, y_0) \in D. \quad \forall (x, y) \in D.$$

$$\begin{aligned} u(x, y) - u(x_0, y_0) &= \int_{(x_0, y_0)}^{(x, y)} Pdx + Qdy = \int_{(x_0, y_0)}^{(x, y_0)} Pdx + Qdy + \int_{(x, y_0)}^{(x, y)} Pdx + Qdy \\ &= \int_{x_0}^x P(v, y_0) dv + \int_{y_0}^y Q(x, w) dw \end{aligned}$$

$$\therefore u(x, y) = \int_{x_0}^x P(v, y_0) dv + \int_{y_0}^y Q(x, w) dw + c.$$

2°. 不定积分法

$$du = Pdx + Qdy$$

$$\frac{\partial u}{\partial x} = P(x, y)$$

$$\frac{\partial u}{\partial y} = Q.$$

$$u_1(x, y) = \int P(x, y) dx$$

$$\frac{\partial u_1}{\partial x} = P(x, y)$$

$$\frac{\partial(u - u_1)}{\partial x} = 0$$

$$\text{令 } u(x, y) = \underline{u_1(x, y)} + \underline{c(y)}$$

$$Q = \frac{\partial u}{\partial y} = \frac{\partial u_1}{\partial y} + \underline{c'(y)}$$

3°. 凑全微分.

$$Pdx + Qdy$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$$\underline{p dx + q dy}$$

$$\stackrel{p}{=} (x^4 + 4xy^3) dx + \stackrel{q}{=} (6x^2y^2 + 5y^4) dy$$

$$= \underline{x^4 dx} + \underline{(4xy^3 dx + 6x^2y^2 dy)} + \underline{5y^4 dy}$$

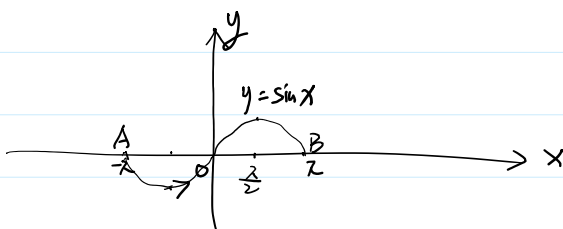
$$= d\left(\frac{x^5}{5}\right) + d(2x^2y^3) + d(y^5) = d\left(\frac{1}{5}x^5 + 2x^2y^3 + y^5\right)$$

$$u(x, y) = \frac{1}{5}x^5 + 2x^2y^3 + y^5 + c.$$

例 4. 计算 $I = \int_L (e^{-x^2} \sin x + 3y - \cos y) dx + (x \sin y - y^4) dy$

L : 从 $A(-\pi, 0)$ 沿 $y = \sin x$ 到 $B(\pi, 0)$ 一段弧.

解:



$$P(x, y) = e^{-x^2} \sin x + 3y - \cos y.$$

$$\overline{AB} = \begin{cases} y = 0 \\ x: -\pi \rightarrow \pi \end{cases}$$

$$Q(x, y) = x \sin y - y^4$$

$$\frac{\partial P}{\partial y} = 3 + \sin y$$

$$\frac{\partial Q}{\partial x} = \sin y$$

$$P_1(x, y) = e^{-x^2} \sin x - \cos y$$

$$\frac{\partial P_1}{\partial y} = \frac{\partial Q}{\partial x}$$

$$I = \int_L P_1(x, y) dx + Q dy + \int_L 3y dx$$

$$P_1(x, 0) = e^{-x^2} \sin x - 1$$

$$= \int_{\overline{AB}} P_1(x, y) dx + Q dy + \int_L 3y dx$$

$$= \int_{-\pi}^{\pi} P_1(x, 0) dx + \int_{-\pi}^{\pi} 3 \sin x dx$$

$$= \int_{-\pi}^{\pi} (e^{-x^2} \sin x - 1) dx = -2\pi.$$

≡ 全微分.

$$\frac{dy}{dx} = f(x, y)$$

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$$\frac{dy}{dx} = f(x, y)$$

$$dy = f(x, y) dx$$

$$P(x, y) dx + Q(x, y) dy = 0.$$

定义3.

若 $P dx + Q dy$ 是 $u(x, y)$ 的全微分. 称 $P(x, y) dx + Q(x, y) dy = 0$ 为全微分方程

$$u(x, y) = C. \text{ 为 } P dx + Q dy = 0 \text{ 隐式解.}$$

例6. 求 $(\ln y - \frac{y}{x}) dx + (\frac{x}{y} - \ln x) dy = 0$ 通解.

解: $P(x, y) = \ln y - \frac{y}{x}$ $Q(x, y) = \frac{x}{y} - \ln x$

$$\frac{\partial P}{\partial y} = \frac{1}{y} - \frac{1}{x} = \frac{\partial Q}{\partial x}$$

$$\begin{aligned} P dx + Q dy &= \ln y dx - \frac{y}{x} dx + \frac{x}{y} dy - \ln x dy \\ &= (\ln y dx + \frac{x}{y} dy) - (\frac{y}{x} dx + \ln x dy) \\ &= d(x \ln y) - d(y \ln x) \end{aligned}$$

$$\therefore P dx + Q dy = 0 \text{ 通解 } \underline{x \ln y - y \ln x = C.}$$

$$P dx + Q dy = 0.$$

$$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$$

$u(x, y) \neq 0$ $u(x, y) P dx + u(x, y) Q dy = 0$ 全微分方程

称 $u(x, y)$ 为 $P dx + Q dy = 0$ 积分因子.

$$\frac{\partial(uP)}{\partial y} = \frac{\partial(uQ)}{\partial x}$$

$$u'_y P + u P'_y = u'_x Q + u Q'_x$$

$$\mu_y' p + \mu p_y' = \mu_x' q + \mu q_x'$$